Introduction to Systems

Outlines

- Basics of Systems
- Types of Systems
- Applications of Systems

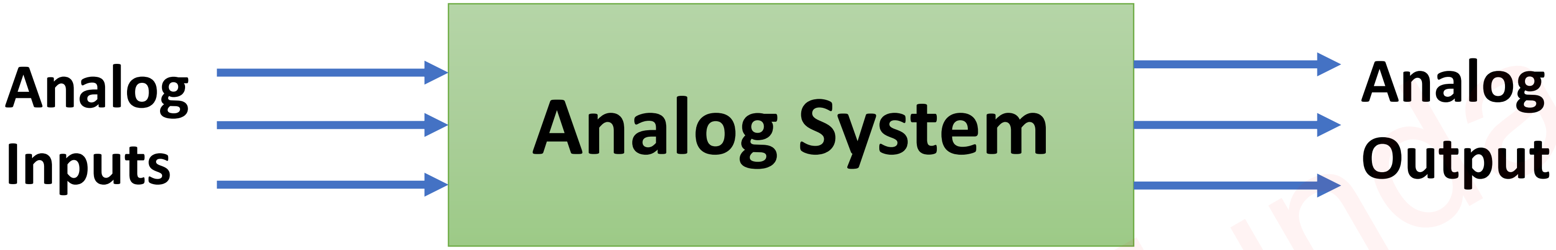
Basics of Systems

- System is a device that performs operations on signals and it accomplishes a function to produce desired output.

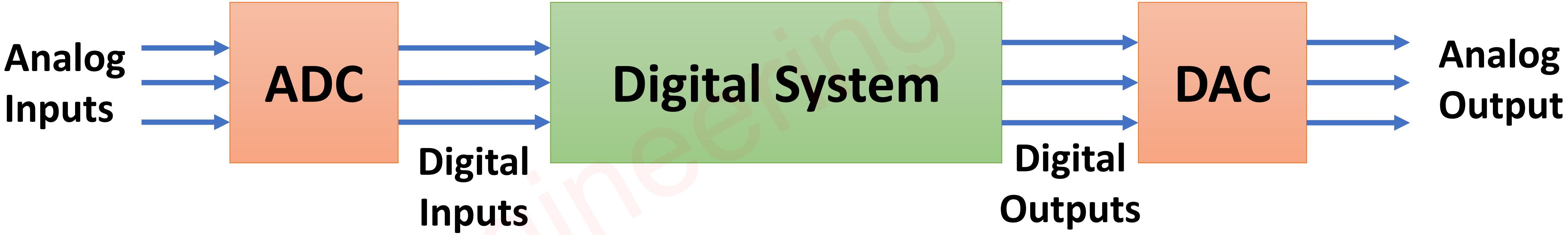
- Elements of Systems:
  - Inputs {Analog or Digital Signal based on Input Sensors}
  - System Process {Can be Analyzed by Transfer Function}
  - Output {Analog or Digital Signal based on Output Actuators}
- Examples:
  - Electrical Systems {Amplifiers, Filters, Circuits etc.}
  - Mechanical Systems {Engines, Robots, Hydraulic systems etc.}
  - Digital Systems {Computers, Smart Watches etc.}

Types of Systems

1. Analog Systems



2. Digital Systems



- ❑ Most information signals are analog in nature.
- ❑ Most widely used systems are Digital Systems {Hence, we need ADC and DAC with Digital Systems}.
- ❑ Analog Signals needs Infinite Memory for storage while Digital Signals can be easily stored in Memory.

Applications of Systems

- ❑ Biomedical Systems: {MRI, CT Scans, ECG etc.}
- ❑ Electrical Systems: {TV, Audio-Systems etc.}
- ❑ Mechanical Systems: {CNC, Engines etc.}

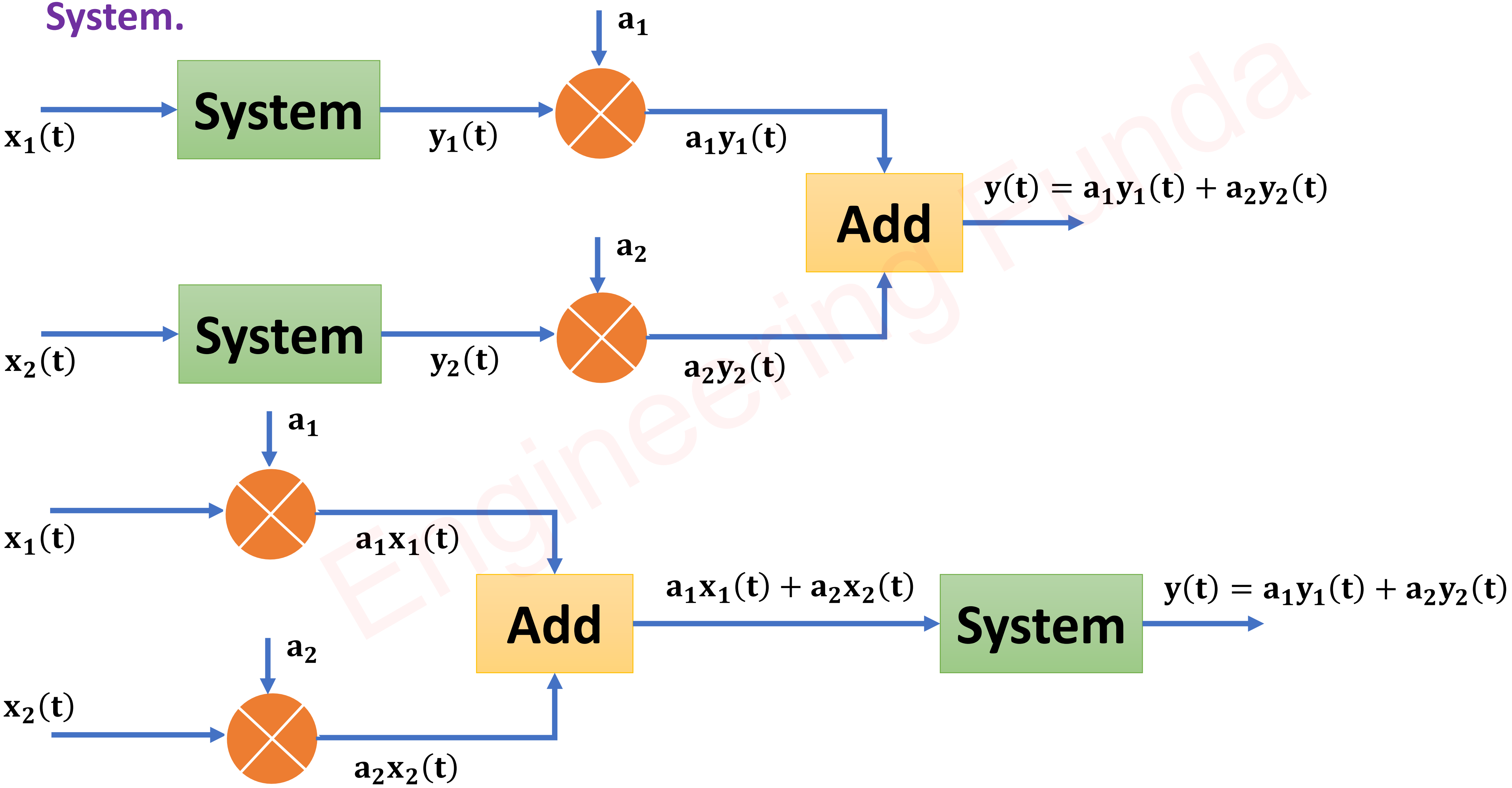
Classifications of Systems

Systems can be Classified in following ways:

- Linear and Non-Linear System
- Time Variant and Timer Invariant System
- Linear Time Invariant and Linear Time Variant System
- Static and Dynamic System
- Causal and Noncausal System
- Invertible and Noninvertible System
- Stable and Unstable System

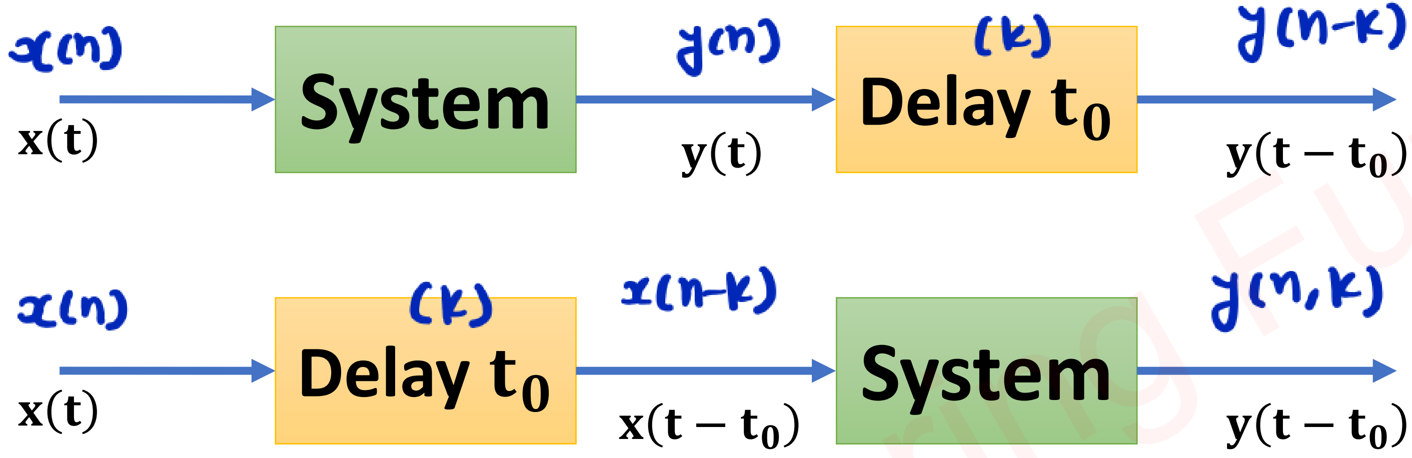
Linear and Non-Linear System

- If The System follows the Super Position Theorem, then the System is said to be a Linear System.



Time Invariant and Time Variant System

- A System is said to be a Time Invariant system if a time shift in the input signal results in a corresponding time shift in the output.



$\Rightarrow y(n-k) = y(n,k)$

– Time Invariant system.

Linear Time Invariant and Linear Time Variant System

- If the System is Linear and Time-Invariant, then the system is a Linear Time Invariant (LTI) System.

Static and Dynamic System

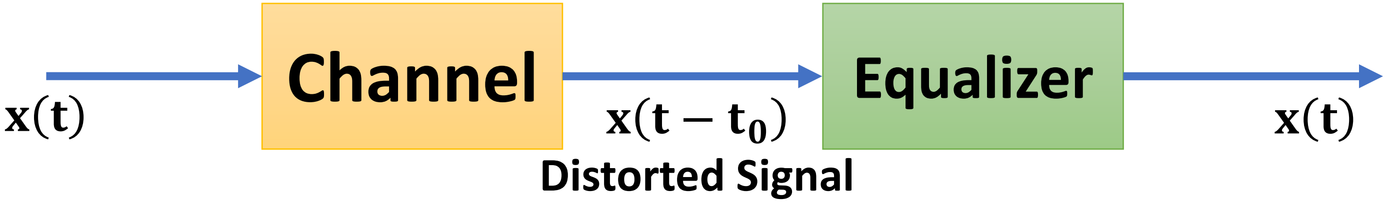
- Static Systems are Memoryless Systems. The output of static systems depends only on present inputs.
- Dynamic Systems have Memory within them, so the Output of Dynamic systems depends on present, past, and future inputs.

Causal and Noncausal System

- A system is "Causal" if its output depends only on the present and past input values and does not depend on its future inputs.
- Noncausal System depends on future inputs.

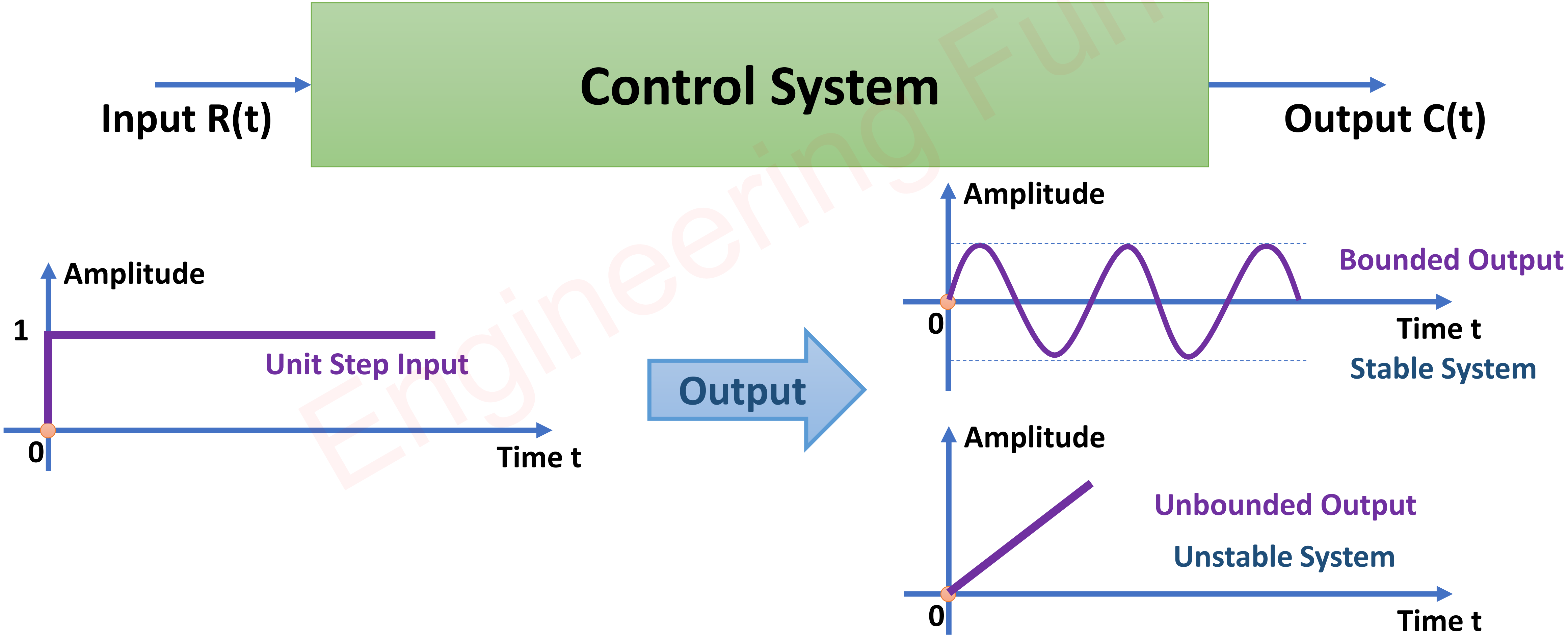
Invertible and Noninvertible System

- A system is "Invertible" if the Input of the system can be recovered from the system output.
- An Equalizer can be used to recover Distorted Signal.



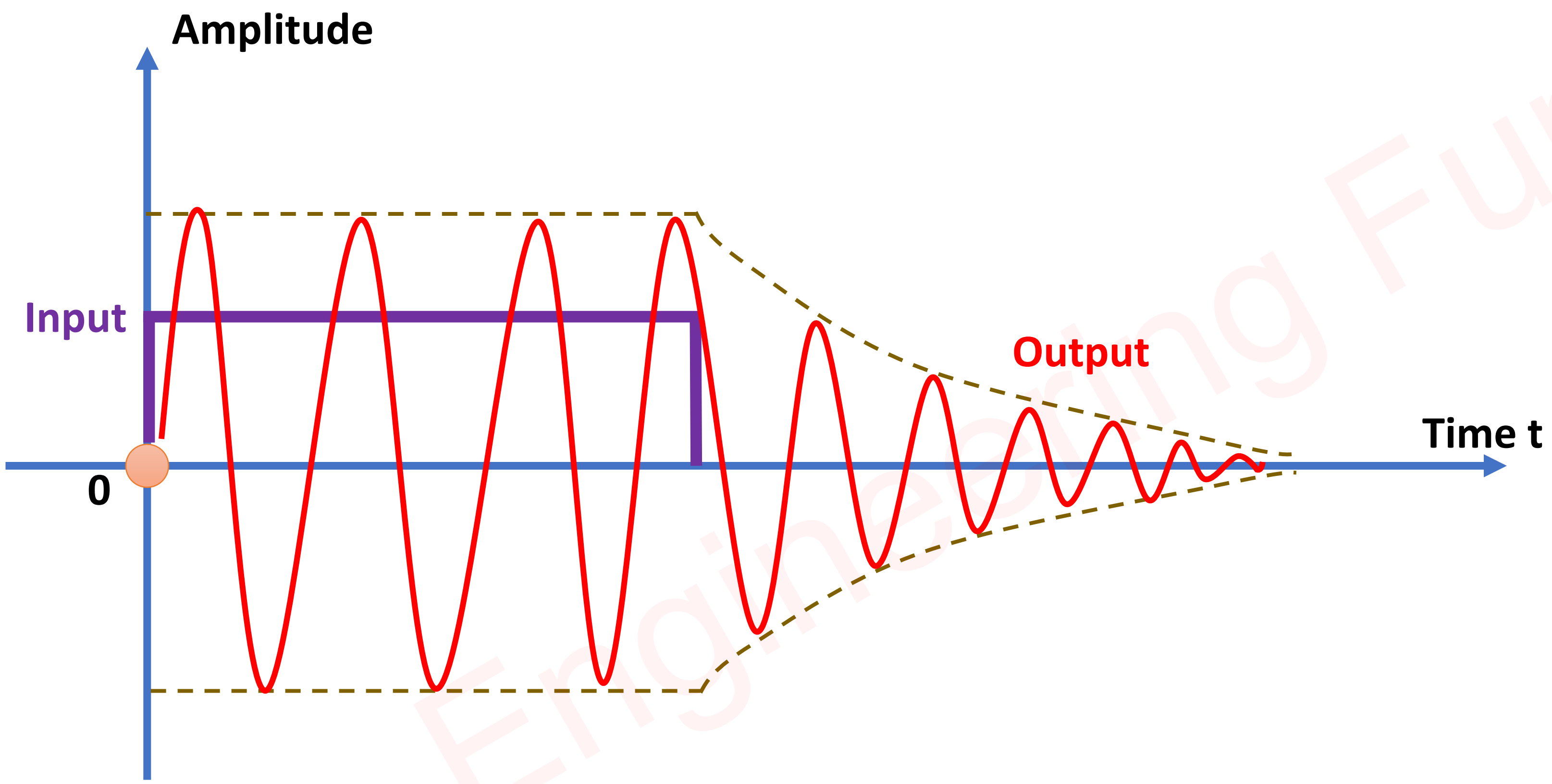
Stable and Unstable System

- If the system produces BIBO {Bounded Input to Bounded Output} then the system is said to be Stable System.
- If Transfer Function is given then from the location of Poles, stability can be identified. If any pole is in RHP {Right Half Plane} then the System is unstable.



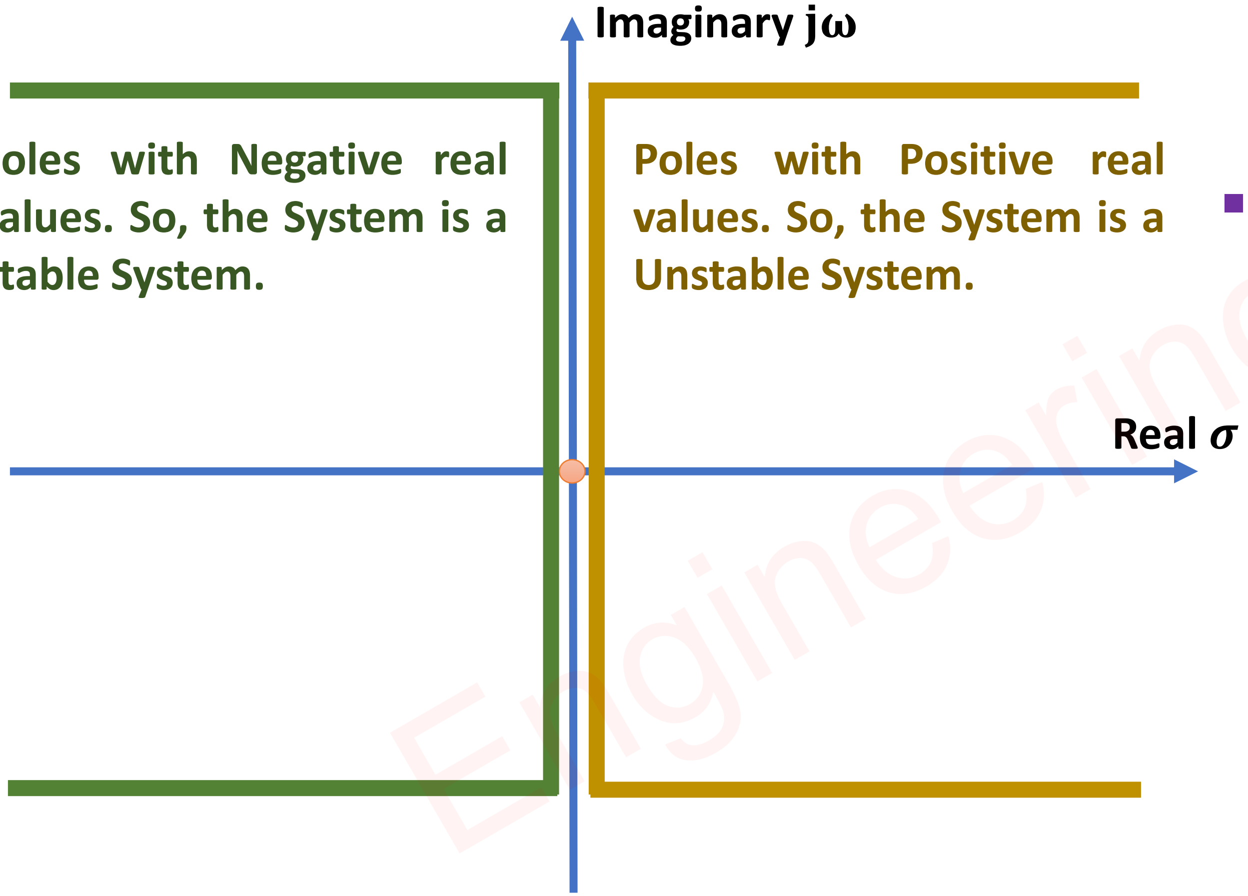
2nd Stability Criteria for System

- A system is Asymptotically stable if The output tends towards zero in the absence of Input. Irrespective of initial conditions.



3rd Stability Criteria for System

- **The stability of the System depends on Poles. If all the poles are located in the left half of the S-Plane, then the system is a Stable System.**



For Poles with Zero real values, System is Marginally Stable.

$$\Rightarrow \mathbf{T(s)} = \frac{\mathbf{C(S)}}{\mathbf{R(S)}} = \frac{\mathbf{N(S)}}{\mathbf{D(S)}}$$

- **N(S) roots gives Zeros and D(S) roots gives Poles.**

Linear & Nonlinear Systems

Question 1 – Check whether the given system is linear or not, $y(t) = e^{x(t)}$.

- If Input is $x_1(t)$

  $\Rightarrow y_1(t) = e^{x_1(t)}$

- If Input is $x_2(t)$

  $\Rightarrow y_2(t) = e^{x_2(t)}$

- LHS $= a_1 y_1(t) + a_2 y_2(t)$

  $= a_1 e^{x_1(t)} + a_2 e^{x_2(t)}$

- If Input is $a_1 x_1(t) + a_2 x_2(t)$

- RHS $= e^{(a_1 x_1(t) + a_2 x_2(t))}$

- LHS $\neq$ RHS

- So, System is non-linear System.

Question 2 – **Check whether the given system is linear or not, $y(t) = x^2(t)$.**

~ If Input is $x_1(t)$

$\Rightarrow y_1(t) = x_1^2(t)$

~ If Input is $x_2(t)$

$\Rightarrow y_2(t) = x_2^2(t)$

~ LHS $= a_1 y_1(t) + a_2 y_2(t)$

$= a_1 x_1^2(t) + a_2 x_2^2(t)$

~ If Input is $a_1 x_1(t) + a_2 x_2(t)$

~ RHS $= (a_1 x_1(t) + a_2 x_2(t))^2$

~ LHS $\neq$ RHS

~ So, System is non-linear System.

Question 3 – **Check whether the given system is linear or not, $y(n) = x(n) + nx(n+1)$.**

~ If Input is $x_1(n)$

$\Rightarrow y_1(n) = x_1(n) + n x_1(n+1)$

~ If Input is $x_2(n)$

$\Rightarrow y_2(n) = x_2(n) + n x_2(n+1)$

~ LHS $= a_1 y_1(n) + a_2 y_2(n)$

$= a_1(x_1(n) + n x_1(n+1)) + a_2(x_2(n) + n x_2(n+1))$

~ If Input is $a_1 x_1(n) + a_2 x_2(n)$

~ RHS $= a_1 x_1(n) + a_2 x_2(n) + n a_1 x_1(n+1) + n a_2 x_2(n+1)$

~ LHS = RHS

~ So, System is linear System.

Time Variant & Time Invariant Systems

Question 1 – Check whether the given system is Time Invariant or not, $y(n) = nx(n) + x(n-2)$.

→ $y(n) = n x(n) + x(n-2)$

→ $y(n-k) = (n-k)\, x(n-k) + x(n-k-2)$

→ $y(n, k) = n\, x(n-k) + x(n-k-2)$

→ $y(n-k) \neq y(n, k)$

→ So, This is Time Variant System.

Question 2 – Check whether the given system is Time Invariant or not, $y(t) = x(t) - 3u(t)$.

→ $y(t-k) = x(t-k) - 3u(t-k)$

→ $y(t,k) = x(t-k) - 3u(t)$

→ $y(t-k) \neq y(t,k)$

→ so, This is Time Variant System.

Question 3 – Check whether the given system is Time Invariant or not,
$\mathbf{y(n) = x(n) - x(n-3)}$.

~ $y(n-k) = x(n-k) - x(n-k-3)$

~ $y(n,k) = x(n-k) - x(n-k-3)$

~ $y(n-k) = y(n,k)$

~ so, This is time Invariant system.

Question 4 – **Check whether the given system is Time Invariant or not, $y(n) = x^2(n) - 2n$.**

$\rightsquigarrow$ $y(n-k) = x^2(n-k) - 2(n-k)$.

$\rightsquigarrow$ $y(n,k) = x^2(n-k) - 2n$

$\rightsquigarrow$ $y(n-k) \neq y(n,k)$

$\rightsquigarrow$ so, This is Time Variant System.

LTI & LTV Systems

Question 1 – Check whether the given system is LTI or LTV or None, $y(n) = nx(n)$.

→ Linearity Check.

- If Input is $x_1(n)$.

$\Rightarrow y_1(n) = n x_1(n)$

- If Input is $x_2(n)$

$\Rightarrow y_2(n) = n x_2(n)$

- $LHS = a_1 y_1(n) + a_2 y_2(n)$

$= a_1 n x_1(n) + a_2 n x_2(n)$

- If Input is $a_1 x_1(n) + a_2 x_2(n)$

- $RHS = n a_1 x_1(n) + n a_2 x_2(n)$

- $LHS = RHS$

- So, System is linear System.

→ Time Variance Check

- $y(n-k) = (n-k)\, x(n-k)$

- $y(n,k) = n\, x(n-k)$

- $y(n-k) \neq y(n,k)$

- So, System is time Variance system.

- Hence, System is LTV System.

Question 2 – Check whether the given system is LTI or LTV or None, $y(n) = x(n) + x(n-1)$.

→ Linearity Check

- If input is $x_1(n)$.
  - $y_1(n) = x_1(n) + x_1(n-1)$
- If input is $x_2(n)$
  - $y_2(n) = x_2(n) + x_2(n-1)$

→ $LHS = a_1 y_1(n) + a_2 y_2(n)$

$= a_1(x_1(n) + x_1(n-1)) + a_2(x_2(n) + x_2(n-1))$

- If input is $a_1 x_1(n) + a_2 x_2(n)$

→ $RHS = a_1 x_1(n) + a_2 x_2(n) + a_1 x_1(n-1) + a_2 x_2(n-1)$

- LHS = RHS
- So, system is linear system.

→ Time Variance Check

- $y(n-k) = x(n-k) + x(n-k-1)$
- $y(n,k) = x(n-k) + x(n-k-1)$
- $y(n-k) = y(n,k)$
- So, system is time invariance.
- Hence, system is LTI system.

Static & Dynamic Systems

Question 1 – Check whether the given system is Static or Dynamic, $y(n) = nx(n)$.

Memory less. Present I/p

Question 2 – Check whether the given system is Static or Dynamic, $y(n) = x^2(n) + x(n-2)$.

Memory Needed. Present I/p Past I/p

Question 3 – Check whether the given system is Static or Dynamic, $y(n) = x(n) + u(n+2)$.

Memory less Present I/p Fixed Signal.

Question 4 – Check whether the given system is Static or Dynamic, $y(n) = x(2n)$.

$$\left.\begin{array}{l} -\ y(1) = x(2) \\ -\ y(2) = x(4) \end{array}\right\} \text{Future I/p.}$$

Question 5 – Check whether the given system is Static or Dynamic, $y(n) = e^{x(n)}$.

Memory less Present I/p

Causal & Non-Causal Systems

Check whether the given systems are Causal or Non-Causal,

1. $$y(n) = \underbrace{nx(n)}_{\text{Present I/P}} + \underbrace{x(n-2)}_{\text{Past I/P}}.$$

→ Causal system.

2. $$y(t) = \underbrace{x^2(t)}_{\text{Present I/P}} + \underbrace{x(t+1)}_{\text{Future I/P}}.$$

→ Non-Causal system.

3. $$y(n) = \underbrace{x^2(n)}_{\text{Present I/P}} + \underbrace{x(n-1)}_{\text{Past I/P}} + \underbrace{x(n+1)}_{\text{Future I/P}}.$$

→ Non-Causal system

4. $$y(n) = x(2n).$$

$$\left.\begin{array}{l} - \ y(1) = x(2) \\ - \ y(2) = x(4) \end{array}\right\} \text{Future I/P}$$

→ Non-Causal system.

5. $y(n) = e^{x(n)}$.

Present I/P

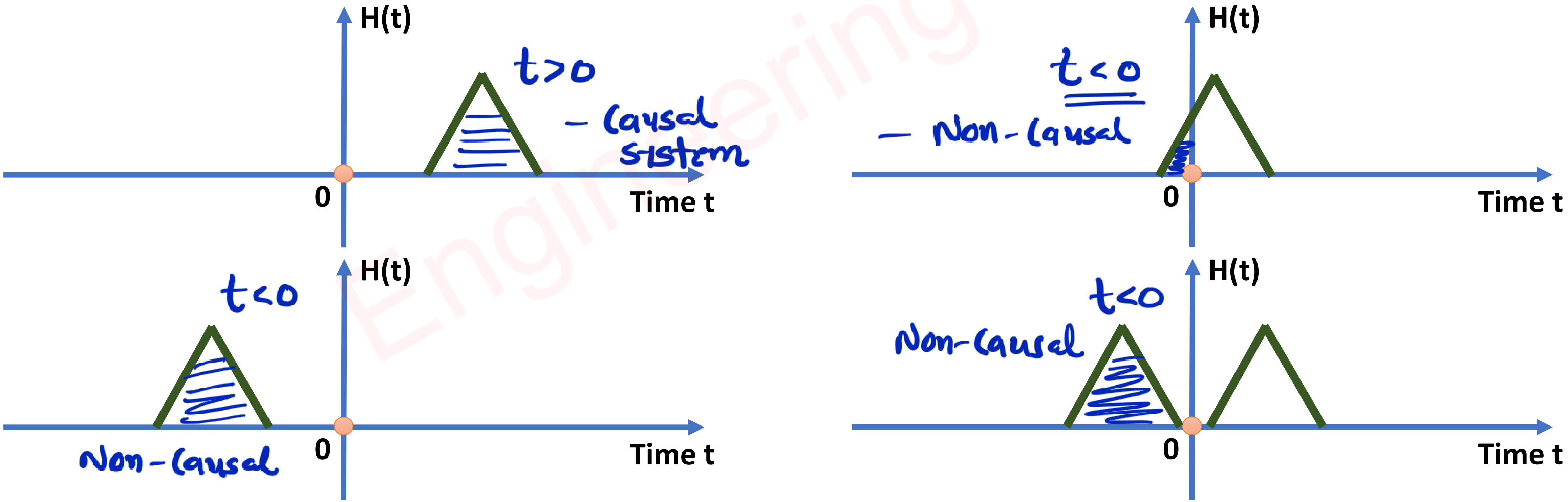
→ causal system.

6. If excitation is applied to a system at time $t = T$ and its response is zero for $-\infty < t < T$.

O/P appear after I/P, means o/p depends on present & past I/P.

– So, system is causal system.

7. Check Causality for the following systems. {H(t) is the time response of systems}



8. $\mathbf{y(n) = x(-n).}$

→ $n>0$, O/P $y(n)$ depends on Past I/P.
→ $n<0$, O/P $y(n)$ depends on future I/P.

Non-Causal system.

9. $\mathbf{y(t) = \int_{-2}^{2t} x(m)dm.}$

- For 2t limits, O/P depends on future I/P.

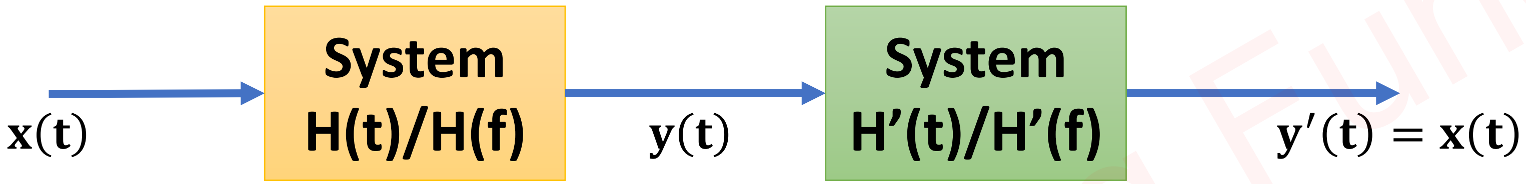
- Non-Causal system.

- All Static Systems are Causal Systems but All Causal Systems may not be Static Systems.
- All Non-Causal Systems are Dynamic Systems but All Dynamic Systems may not be Non-Causal Systems.
- If the System has differentiation/ Integration of Input/ Output, then the system is a Dynamic System.

Invertible & Non-Invertible Systems

Basics of Invertible & Non-Invertible Systems

- A system is "Invertible" if the Input of the system can be recovered from the system output.



- Response of H(t)

$$\Rightarrow \mathbf{y(t) = x(t) * h(t)}$$

- In Frequency Domain

$$\Rightarrow \mathbf{y(f) = x(f) \times h(f)}$$

- Response of H'(t)

$$\Rightarrow \mathbf{y'(t) = x(t) * h(t) * h'(t)}$$

- In Frequency Domain

$$\Rightarrow \mathbf{y'(f) = x(f) \times h(f) \times h'(f)}$$

- If h(f) x H'(f) = 1

$$\Rightarrow \mathbf{y'(f) = x(f)}$$

- If we take inverse transform

$$\Rightarrow \mathbf{y(t) = x(t)}$$

Difference between Invertible & Non-Invertible Systems

Property	Invertible	Non-Invertible
Input Recovery	Possible	Not Possible
Uniqueness	Unique mapping between input and output	Non-unique mapping between input and output
Information Loss	No Loss	Loss Occurs

Importance in Applications

- **Invertible Systems:** These are used in applications where recovering the original signal is critical, such as communication systems, encryption, and data compression.
- **Non-Invertible Systems:** These are common in irreversible processes like image processing (quantization) or audio clipping, where exact recovery is not needed or possible.

Examples of Non-Invertible Systems

- **Squaring:** $y(t) = x^2(t)$, For $x(t) = +\sqrt{y(t)}$ and $x(t) = -\sqrt{y(t)}$, we have same Output.
- **Clipping:** Clipped Signal can not be recovered.
- **One to Many Mapping / Overlapping.**

Stable & Unstable Systems

Check whether the given systems are Stable or Unstable,

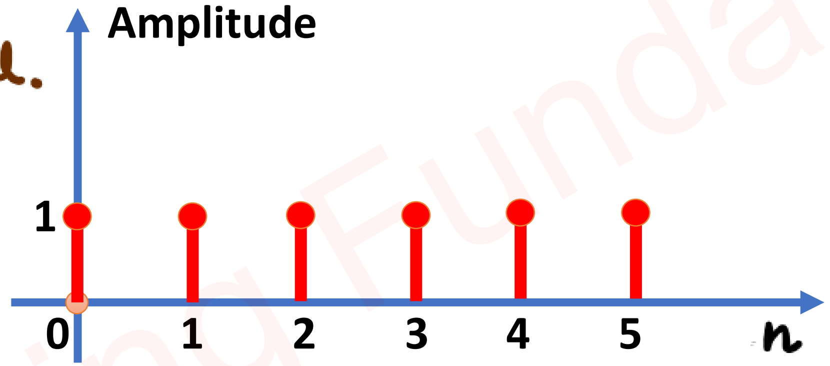
1. $y(n) = x^2(n)$.

⤳ Standard bounded Input is Unit step Signal.

- $y(n) = u^2(n)$

  $= u(n)$

- O/p is bounded. Stable system.



2. $y(t) = 2x^2(t) + x(t)$.

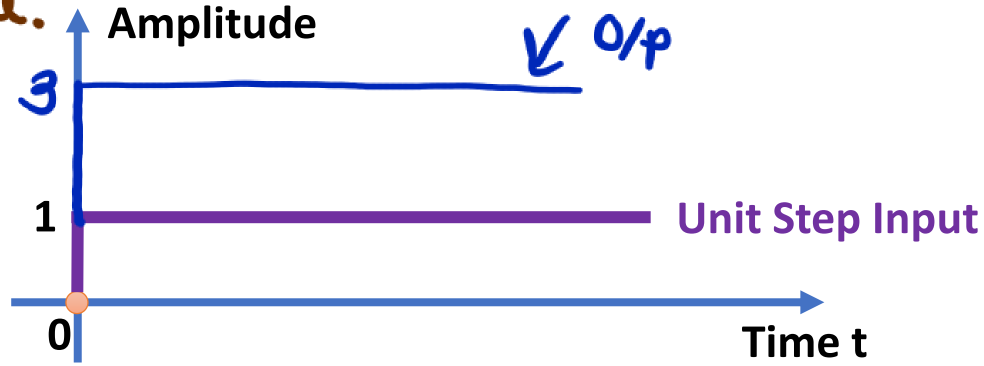
⤳ Standard bounded Input is Unit step Signal.

- $y(t) = 2u^2(t) + u(t)$

  $= 2u(t) + u(t)$

  $= 3u(t)$

- O/p is bounded, Stable System.



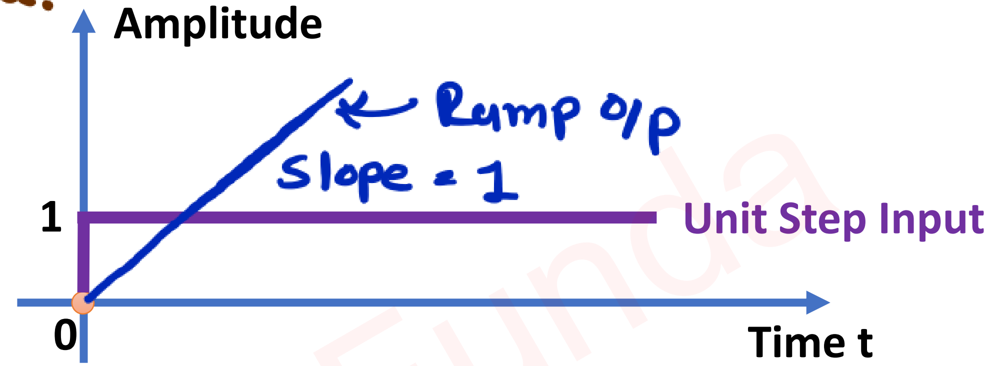
3. $y(t) = \int x(t)dt$

↝ Standard bounded Input is Unit step Signal.

- $y(t) = \int u(t)\, dt$

  $= r(t)$

- O/p is unbounded, Unstable System.



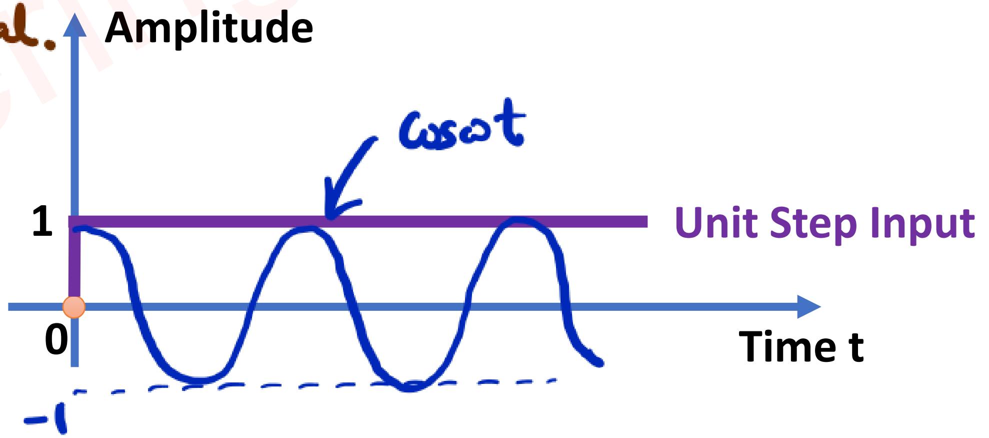
4. $y(t) = x(t) \cos \omega t.$

↝ Standard bounded Input is Unit step Signal.

- $y(t) = u(t) \cos \omega t$

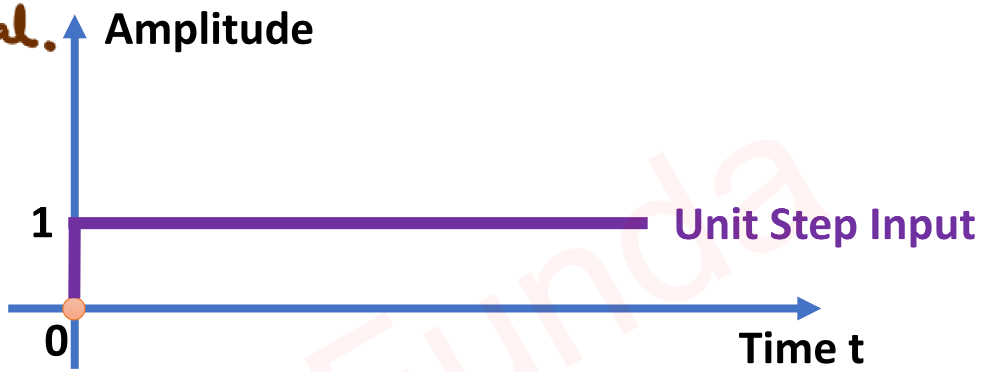
  $= \cos \omega t \;, \; t \geq 0$

- O/p is bounded, stable System.



5. $y(t) = \frac{dx(t)}{dt}$

⤳ Standard bounded Input is Unit step Signal.



- $y(t) = \frac{du(t)}{dt}$

$= \text{Impulse.}$

- O/p is unbounded, Unstable System.

6. $\frac{y(S)}{x(S)} = \frac{k(S^2+4S+1)}{S^3+2S^2+6S}$ ← Roots (Zeros) ← Roots (Poles)

⤳ stability depends on poles.

- $S = 0$, - $S^2 + 2S + 6 = 0$

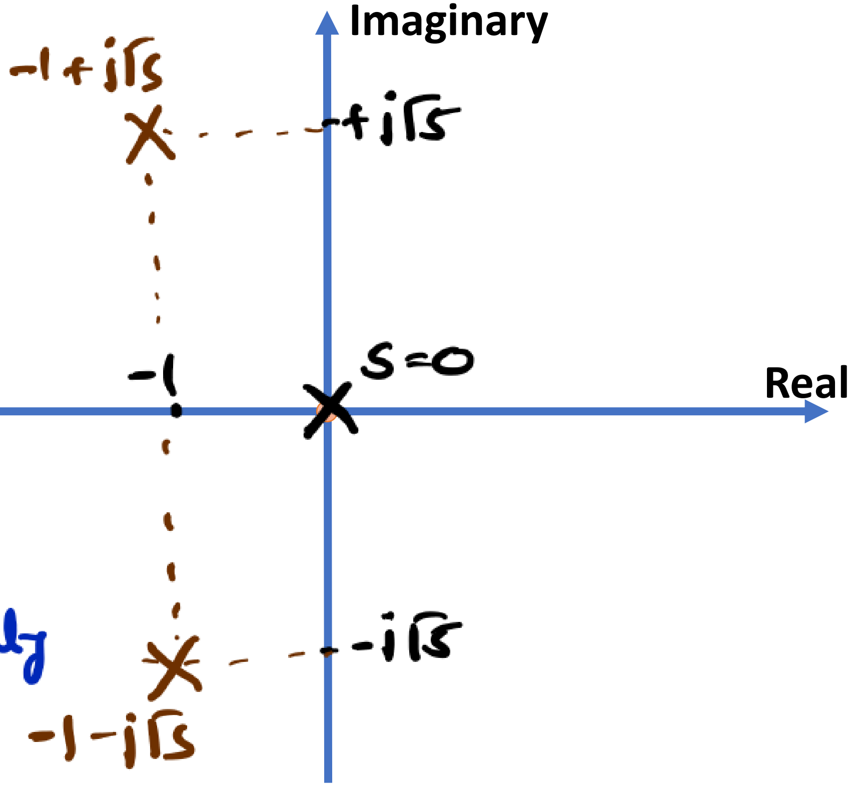
$\Rightarrow (S+1)^2 + 5 = 0$

$\Rightarrow (S+1)^2 = -5$

$\Rightarrow (S+1) = \pm j\sqrt{5}$

$\Rightarrow S = -1 \pm j\sqrt{5}$

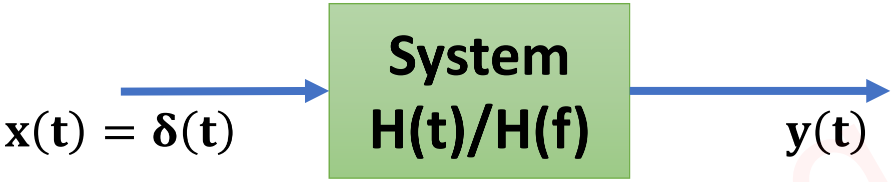
→ As pole is there on Imag axis, we have marginally stable system.



Impulse Response of Systems

Basics of Impulse Response of Systems

- A system's impulse response is the system's output when an impulse function (a Dirac delta function, δ(t)) is applied as an input.



- In the Continuous Time Domain, It is given by:

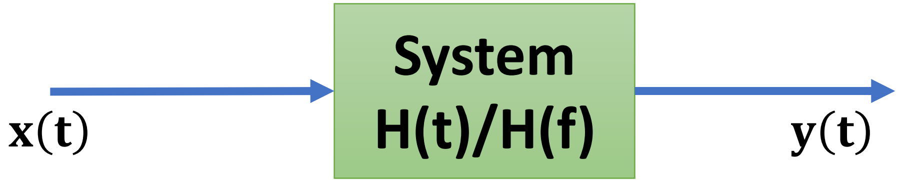
$$\Rightarrow \mathbf{h(t) = H[\delta(t)]}$$

- In the Discrete Time Domain, It is given by:

$$\Rightarrow \mathbf{h(n) = H[\delta(n)]}$$

- Impulse Response is used to Analyse different Types of Systems {Control Systems, Signal Processing, Mechanical Systems, and Communication Systems}.
- Impulse Response is a Complete and Practical Tool for LTI Systems and It doesn't give complete details of Non LTI Systems.

Key Parameters of Impulse Response of Systems



❑ Output of the System:

$$\Rightarrow y(t) = x(t) * h(t)$$

$$\Rightarrow y(f) = x(f) \times h(f)$$

❑ For Causal System:

$$\Rightarrow h(t) = 0, t < 0$$

❑ For Stable System:

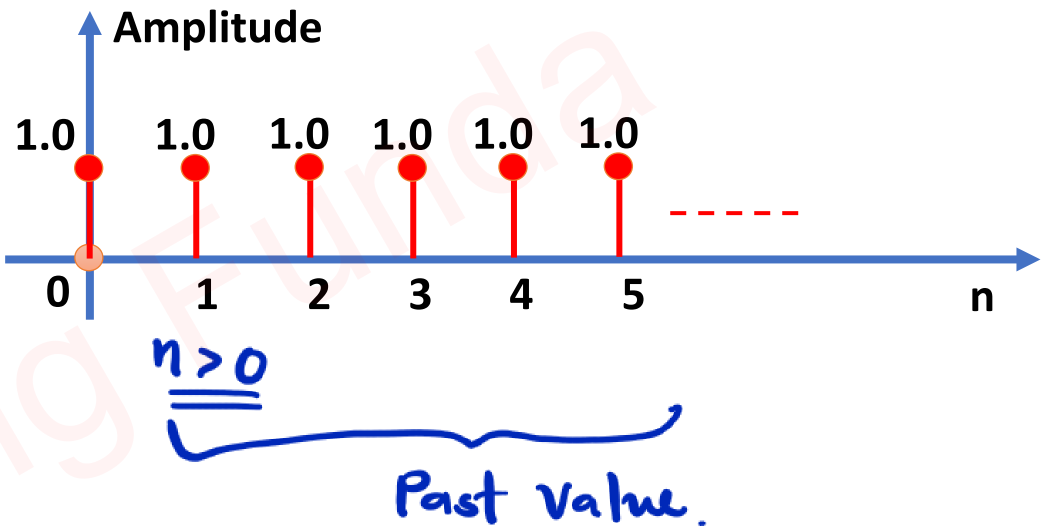
$$\Rightarrow \int_{-\infty}^{\infty} h(t)dt < \infty$$

❑ For Static System:

$$\Rightarrow h(t) = 0, t < 0 \text{ and } t > 0$$

Example 1. For the given impulse response, determine whether system is Static/Dynamic, Causal/Non-Causal, and Stable/Unstable.

$$\Rightarrow h(n) = u(n)$$



~ $h(n) \neq 0, \; n > 0$

- So, system is dynamic system.

~ $h(n) = 0, \; n < 0$

- So, system is causal system.

~ $\int_{-\infty}^{\infty} h(n)\, dn = \int_{0}^{\infty} 1\, dn$

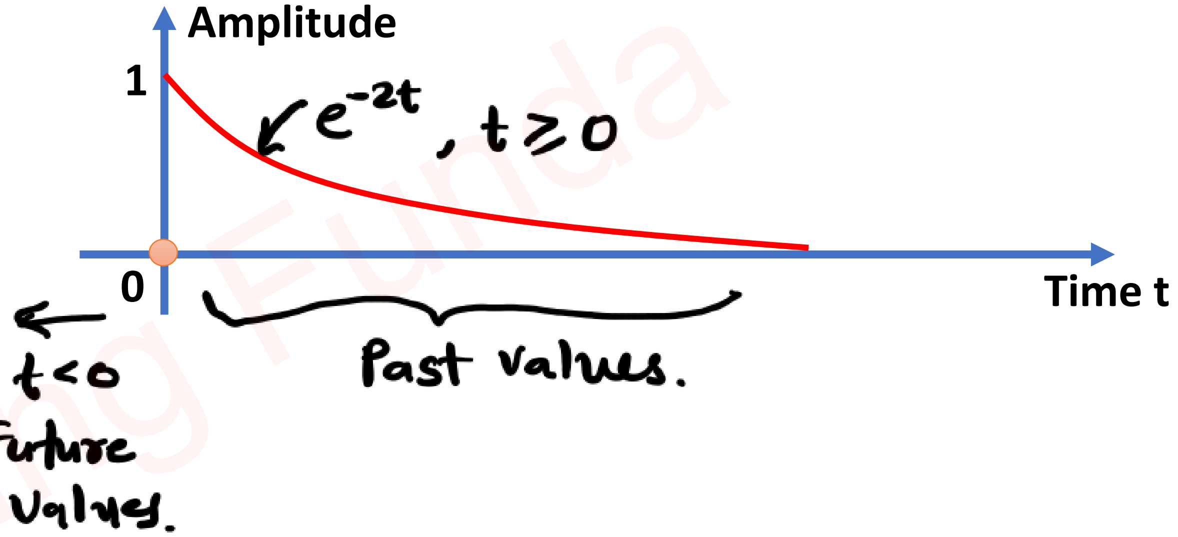
$= [n]_0^{\infty}$

$= \infty$

- So, given system is unstable system.

Example 2. **For the given impulse response, determine whether system is Static/Dynamic, Causal/Non-Causal, and Stable/Unstable.**

$$\Rightarrow h(t) = e^{-2t}u(t)$$



~ $h(t) \neq 0$, $t > 0$

- So, system is dynamic system.

~ $h(t) = 0$, $t < 0$

- So, System is causal system.

~ $\int_{-\infty}^{\infty} h(t)\,dt = \int_{0}^{\infty} e^{-2t}\,dt$

$= \left[\frac{e^{-2t}}{-2}\right]_0^{\infty}$

$= -\frac{1}{2}\left[e^{-\infty} - e^{0}\right]$

$= 1/2 \neq \infty$

- So, System is stable system.

Example 3. **For the given impulse response, determine whether system is Static/Dynamic, Causal/Non-Causal, and Stable/Unstable.**

$$\Rightarrow \mathbf{h(t) = e^{2t}u(-1-t)}$$

~ $h(t) \neq 0, \quad t \leq -1$

– So, System is dynamic system.

~ $h(t) \neq 0, \quad t < 0$

– So, System is non-causal system.

~ $\int_{-\infty}^{\infty} h(t)\, dt = \int_{-\infty}^{-1} e^{2t}\, dt$

$$= \left[\frac{e^{2t}}{2}\right]_{-\infty}^{-1}$$

$$= \frac{1}{2}\left[e^{-2} - e^{-\infty}\right]$$

$$= \frac{1}{2e^{2}} \neq \infty$$

– Stable System.